

Supply Chain Analytics – Business Insights Report

This report summarizes business insights derived from SQL-based analysis of a supply chain dataset. The objective is to demonstrate how data can support operational efficiency, supplier performance management, logistics optimization, pricing strategy, and revenue growth initiatives.

1. Revenue by Product Type

Business Question: Which product categories generate the most revenue?

Business Insight: Insight: Identifies the highest-performing product categories and helps management prioritize inventory, marketing budgets, procurement planning, and growth investments.

2. Defect Rate Analysis

Business Question: Which SKUs have defect rates above acceptable thresholds?

Business Insight: Insight: Highlights quality risks, potential customer satisfaction issues, warranty exposure, and supplier quality concerns requiring corrective action.

3. Supplier Lead Time Performance

Business Question: Which suppliers have the highest average lead times?

Business Insight: Insight: Helps identify suppliers contributing to procurement delays and inventory shortages. Supports supplier evaluation and negotiation decisions.

4. Shipping Cost by Carrier

Business Question: Which shipping carriers contribute the highest logistics costs?

Business Insight: Insight: Enables carrier cost benchmarking, contract optimization, and transportation cost reduction initiatives.

5. Below-Average Price SKU Analysis

Business Question: Which SKUs are priced below their category average?

Business Insight: Insight: Supports pricing strategy reviews, profitability analysis, and opportunities for margin improvement.

6. Transportation Mode Utilization

Business Question: What percentage of shipments use Road, Air, Rail, or Sea transport?

Business Insight: Insight: Reveals transportation dependency patterns and opportunities to optimize cost, speed, and sustainability.

7. Top Revenue Generating SKUs

Business Question: Which products contribute the highest revenue?

Business Insight: Insight: Identifies strategic products driving business performance and supports inventory prioritization and demand planning.

8. Lead Time Gap Analysis

Business Question: Where is the gap between total lead time and manufacturing lead time unusually high?

Business Insight: Insight: Indicates potential bottlenecks in logistics, warehousing, customs, procurement, or supplier coordination processes.

9. Customer Demographic Revenue Contribution

Business Question: Which demographic segments generate the most revenue?

Business Insight: Insight: Supports targeted marketing, customer segmentation, and customer acquisition strategies.

10. Route-Level Logistics Cost Analysis

Business Question: Which transportation routes have the highest average logistics costs?

Business Insight: Insight: Helps optimize route planning, identify inefficiencies, and improve logistics profitability.

Executive Summary

The SQL analysis demonstrates the practical application of data analytics in supply chain management. The findings can help stakeholders improve supplier performance, reduce logistics costs, enhance product quality, optimize transportation strategies, improve pricing decisions, and focus on high-value customers and products. Key business outcomes include: Revenue growth through product and customer segmentation. Cost optimization through carrier and route analysis. Operational efficiency through lead-time monitoring. Quality improvement through defect tracking. Strategic decision-making through data-driven insights.